
Grounding Image Captions: Reducing Hallucination in Vision-Language Models

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Abstract

Recent advances in multimodal deep learning have enabled vision-language models to generate natural language descriptions of images with impressive fluency. Models such as BLIP (Bootstrapping Language-Image Pre-training) combine visual and textual representations to perform tasks including image captioning, visual question answering and image-text retrieval. However, these models often suffer from hallucination, where generated captions mention objects that do not actually appear in the image. This problem becomes more prominent when models are evaluated on images outside the training distribution.

In this project, we investigate hallucination behavior in the BLIP image captioning model and explore whether selecting captions that are more visually grounded can reduce hallucination errors. We first reproduce the baseline BLIP caption generation system and evaluate hallucination using the CHAIR metric. We then generate multiple candidate captions for each image and implement a mechanism to select the caption that best aligns with the visual content of the image. The goal of this project is to analyze hallucination in vision-language captioning systems and evaluate whether improved caption selection strategies can improve caption reliability, particularly when evaluated on unseen datasets.

1 Introduction and Problem Motivation

Image captioning is a core task in multimodal artificial intelligence that combines computer vision and natural language processing to automatically generate textual descriptions of images. Modern vision-language models such as BLIP have significantly improved caption generation performance by learning joint representations between images and text.

Despite these advances, captioning models frequently generate hallucinated objects, meaning the caption mentions objects that do not exist in the image. This occurs because caption generation models partially rely on statistical patterns learned during training rather than strictly grounding predictions in the visual input.

For example, if an image contains a plate on a table, the model may generate a caption such as “a plate with a fork on the table” even when the image does not contain a fork. Such hallucinations reduce the reliability of captioning systems and can negatively impact applications such as accessibility tools, automated image indexing, and visual understanding systems.

Another important challenge is generalization across datasets. Captioning models are typically trained on datasets such as MS COCO, which contain particular object co-occurrence patterns. When models are evaluated on images from a different dataset, these learned patterns may lead to increased hallucination errors because the model relies on dataset biases rather than visual grounding.

Understanding and reducing hallucination in captioning systems is therefore an important research problem in multimodal deep learning.

2 Project Description and Approach

This project investigates hallucination behavior in vision-language captioning models and explores strategies to improve visual grounding in generated captions. We focus on the BLIP (Bootstrapping Language-Image Pre-training) image captioning model as the baseline system.

First, we reproduce the baseline BLIP captioning pipeline using the official pretrained implementation released by the authors. The model generates captions by learning joint representations between visual features extracted from the image and textual tokens produced by the language model. Using this baseline system, captions will be generated for images from the MS COCO dataset.

To analyze hallucination behavior, the generated captions will be evaluated using the CHAIR (Caption Hallucination Assessment with Image Relevance) metric, which measures how frequently captions mention objects that are not present in the image.

A key objective of this project is to study how hallucination behavior changes when models are evaluated on datasets that differ from their training distribution. Captioning models trained on datasets such as COCO may rely on dataset-specific object co-occurrence patterns. As a result, when the model is evaluated on a different dataset, it may generate captions that include objects that do not exist in the image.

To investigate this phenomenon, the captioning system will be evaluated on both the COCO dataset and an additional dataset such as Flickr30k, which contains images from a different visual distribution. Evaluating across these datasets allows analysis of hallucination behavior across both seen and unseen data distributions.

After establishing the baseline hallucination behavior, the project will explore an improvement to the caption generation process. Instead of selecting only the highest probability caption produced by the model, multiple candidate captions (Top-K captions) will be generated for each image. From this set of candidate captions, a selection mechanism will be implemented to identify the caption that best aligns with the visual content of the image.

The intuition behind this approach is that among multiple generated captions, some captions may be more visually grounded than others. Selecting the caption that better reflects the visual content of the image may therefore reduce hallucinated objects compared to the baseline caption generation method.

3 Datasets

The primary dataset used in this project will be the MS COCO dataset, one of the most widely used datasets for image captioning research. The dataset contains a large collection of images paired with multiple human-written captions and object annotations. These annotations allow hallucinated objects to be detected using the CHAIR metric.

To further evaluate model generalization, the captioning system will also be tested on the Flickr30k dataset, which contains images and captions drawn from a different visual distribution. Using both datasets allows us to study hallucination behavior across both seen and unseen visual distributions.

4 Project Timeline

Before March 5 - Milestone 1 Conduct a literature review on hallucination in image captioning and identify the research problem, datasets, and evaluation metrics.

March 5 - April 2 - Milestone 2 Set up the BLIP captioning model using pretrained weights and implement the baseline caption generation pipeline. Generate captions for images and evaluate hallucination using the CHAIR metric. Analyze baseline results and prepare preliminary findings.

April - End of Semester Implement multiple caption generation using BLIP and develop the caption selection mechanism. Conduct experiments comparing the baseline caption generation and the proposed method.

Final Week of Classes - Milestone 3 Prepare the final NeurIPS-style report, organize code and experiments in a GitHub repository, and present the project results.

References and Links

- **BLIP: Bootstrapping Language-Image Pre-training** <https://arxiv.org/abs/2201.12086>
- **Object Hallucination in Image Captioning** <https://arxiv.org/abs/1809.02156>
- **MS COCO Dataset** <https://cocodataset.org>
- **Flickr30k Dataset** <https://shannon.cs.illinois.edu/DenotationGraph/>